1. $\lambda = \frac{16\mu}{5\rho}\frac{1}{\sqrt{2\pi RT}}$, $K_n = \frac{\lambda}{L}$, $Re = \frac{\rho v l}{\mu}$ $R = \frac{P}{\rho T} =$

$$\lambda = \frac{16\,(12.2\times10^{-6})}{5\times 0.0182}\times\frac{1}{\sqrt{2\pi(2972)154}} = 1.26492\cdot10^{-6}, \overset{(3SF)}{1.26\cdot10^{-6}}\ m$$

Model: $Re = \frac{0.0182\times 1743\times 0.1524}{12.2\times10^{-6}} = 396,272$ $Re = 3.96\cdot10^{5}$

$$K_n = \frac{\lambda}{L} = \frac{1.26\cdot10^{-6}}{0.1524} = 8.2999\cdot10^{-6}$$ $K_n = 8.30\cdot10^{-6}$

Rarefied effects are unimportant.

Probe: $Re = \frac{0.0182\times 1743\times 0.0762\cdot10^{-3}}{12.2\cdot10^{-6}} = 198.14$ $Re = 198$

$$K_n = \frac{\lambda}{L} = \frac{1.26\cdot10^{-6}}{0.0762\cdot10^{-3}} = 0.0165999$$ $K_n = 1.66\cdot10^{-2}$

Rarefied effects may be important depending on desired solution accuracy as we have just entered the transition flow regime.

2. Maxwell's Equations in tensor notation

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \quad \Rightarrow \quad \epsilon_{ijk} \frac{\partial E_k}{\partial x_j} = -\frac{\partial B_i}{\partial t}$$

$$\vec{\nabla} \times \vec{B} = \epsilon_0 \mu_0 \frac{\partial \vec{E}}{\partial t} + \mu_0 \vec{J} \quad \Rightarrow \quad \epsilon_{ijk} \frac{\partial B_k}{\partial x_j} = \epsilon_0 \mu_0 \frac{\partial E_i}{\partial t} + \mu_0 J_i$$

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0} \quad \Rightarrow \quad \frac{\partial E_i}{\partial x_i} = \frac{\rho}{\epsilon_0}$$

$$\vec{\nabla} \cdot \vec{B} = 0 \quad \Rightarrow \quad \frac{\partial B_i}{\partial x_i} = 0$$

3. find $\langle v f \rangle : \langle v^2 f \rangle^{1/2} : v_m$ ← highest likelihood

Maxwell-Boltzmann distribution: $f(v) = 4\pi v^2 \left(\frac{m}{2\pi kT}\right)^{3/2} \exp\left(-\frac{1}{2}\frac{m}{kT}v^2\right)$

→ mean, first moment: $\langle vf \rangle = \int_0^\infty v\, f(v)\, dv = \int_0^\infty dv 4\pi v^3 \left(\frac{m}{2\pi kT}\right)^{3/2} \exp\left(-\frac{1}{2}\frac{m}{kT}v^2\right)$

$\langle vf \rangle = 4\pi \left(\frac{m}{2\pi kT}\right)^{3/2} \int_0^\infty v^3 \exp\left(-\frac{1}{2}mv^2/kT\right) dv$

the integral can be written in the following form: $I_n(\beta) = \int_0^\infty t^n e^{-\beta t^2} dt$, with $n = 3$

and $\beta = \frac{1}{2}\frac{m}{kT}$, so we can consult an integral table to get the result of this integral as $I_3(\beta) = \frac{1}{2\beta^2} = \frac{1}{2\left(\frac{1}{2}\frac{m}{kT}\right)^2} = \frac{2(kT)^2}{m^2}$

$\langle vf \rangle = 4\pi \left(\frac{m}{2\pi kT}\right)^{3/2} \frac{2(kT)^2}{m^2} = 8\pi \cdot \frac{1}{(2\pi)^{3/2}} \frac{(kT)^{1/2}}{m^{1/2}} = \frac{2^3}{2^{3/2}} \cdot \frac{\pi}{\pi^{3/2}} \sqrt{\frac{kT}{m}} = \sqrt{\frac{8}{\pi}}\sqrt{\frac{kT}{m}} = \sqrt{\frac{8kT}{\pi m}} = \langle vf \rangle$

→ rms, square-root of second moment: $\langle v^2 f \rangle^{1/2} = \left(\int_0^\infty 4\pi v^4 \left(\frac{m}{2\pi kT}\right)^{3/2} \exp\left(-\frac{1}{2}\frac{m}{kT}v^2\right) dv\right)^{1/2}$

$\langle v^2 f \rangle^{1/2} = \left[4\pi \left(\frac{m}{2\pi kT}\right)^{3/2} \int_0^\infty v^4 \exp\left(-\frac{1}{2}\frac{m}{kT}v^2\right) dv\right]^{1/2}$, so we can rewrite the integral part of the expression in the form $I_n(\beta) = \int_0^\infty t^n e^{-\beta t^2} dt$ with $n = 4$ and $\beta = \frac{1}{2}\frac{m}{kT}$ and from a table of integrals we see...

$I_4(\beta) = \frac{3\sqrt{\pi}}{8\beta^{5/2}} = \frac{3\sqrt{\pi}}{8}\left(\frac{2kT}{m}\right)^{3/2}$

$\langle v^2 f \rangle^{1/2} = \left[4\pi \left(\frac{m}{2\pi kT}\right)^{3/2} \left(\frac{3\sqrt{\pi}}{8}\right)\left(\frac{2kT}{m}\right)^{5/2}\right]^{1/2}$

$= \left[\frac{\pi \cdot 3 \cdot \sqrt{\pi}}{\pi^{3/2} 2} \left(\frac{m}{2kT}\right)^{3/2} \left(\frac{2kT}{m}\right)^{5/2}\right]^{1/2} = \left[\frac{3}{2}\left(\frac{kT}{m}\right)\right]^{1/2}$

$\langle v^2 f \rangle^{1/2} = \sqrt{\frac{3kT}{m}}$

→ v_m, most probable speed. set $\frac{df(v)}{dv} = 0$

$\frac{df(v)}{dv} = \frac{d}{dv}\left(4\pi v^2 \left(\frac{m}{2\pi kT}\right)^{3/2} \exp\left(-\frac{1}{2}\frac{m}{kT}v^2\right)\right)$

$= 4\pi \left(\frac{m}{2\pi kT}\right)^{3/2} \frac{d}{dv}\left(v^2 \exp\left(-\frac{1}{2}\frac{m}{kT}v^2\right)\right)$, now apply chain rule & product rule

$= 4\pi \left(\frac{m}{2\pi kT}\right)^{3/2} \left[2v \exp\left(-\frac{1}{2}\frac{m}{kT}v^2\right) + v^2\left(-\frac{1}{2}\frac{m}{kT}(2v)\exp\left(-\frac{1}{2}\frac{m}{kT}v^2\right)\right)\right]$

$\frac{df(v)}{dv} = 4\pi \left(\frac{m}{2\pi kT}\right)^{3/2} \left(2v \exp\left(-\frac{1}{2}\frac{m}{kT}v^2\right)\right)\left[1 - \frac{1}{2}\frac{m}{kT}v^2\right]$

we want the most probable speed, so we force $\frac{df(v)}{dv} = 0$

this is true in 3 cases.

case 1: $m=0$, not of interest since particles in our model have finite masses $m>0$.

case 2: $T \to \infty$, not specifically of interest since we want to know the most probable value of v for some given constant temperature.

Case 3: $\left(1 - \frac{1}{2}\frac{M}{KT}v^2\right) = 0$, which has solutions at $v = \pm\sqrt{\frac{2KT}{m}}$

Since this is speed, $v \geq 0$, so the physically relevant solution is $v = \sqrt{\frac{2KT}{m}}$

$\therefore$ the most probable speed is $v_m = \sqrt{\frac{2KT}{m}}$

$\langle vf \rangle : \langle v^2 f \rangle^{1/2} : v_m \Rightarrow \sqrt{\frac{8kT}{\pi m}} : \sqrt{\frac{3kT}{m}} : \sqrt{\frac{2kT}{m}}$

$\langle vf \rangle : \langle v^2 f \rangle^{1/2} : v_m = \sqrt{8/\pi} : \sqrt{3} : \sqrt{2}$

4. since the gas has mass m and is in thermal equilibrium, it has the velocity distribution $f(v) = 4\pi v^2 \left(\frac{m}{2\pi kT}\right)^{3/2} \exp\left(-\frac{1}{2}\frac{m}{kT}v^2\right)$

a) we are given $E = \frac{1}{2}mv^2$ so $v^2 = \frac{2E}{m}$ $v = \left(\frac{2E}{m}\right)^{1/2}$ (since $v \geq 0$)

and $dv = \frac{1}{2}\left(\frac{2E}{m}\right)^{-1/2}\left(\frac{2dE}{m}\right) = \frac{1}{2}\left(\frac{m}{2E}\right)^{1/2}\left(\frac{2}{m}\right)dE = \frac{1}{2}\left(\frac{2}{mE}\right)^{1/2}dE = \left(\frac{1}{2mE}\right)^{1/2}dE$

we want $f(v)\,dv = f_E(E)\,dE$, so if we perform substitution on $f(v)dv$, we get...

$$f(v)dv = 4\pi\left(\frac{2E}{m}\right)\left(\frac{m}{2\pi kT}\right)^{3/2}\exp\left(-\frac{1}{2}\frac{m}{kT}\left(\frac{2E}{m}\right)\right)\left(\frac{1}{2mE}\right)^{1/2}dE$$

then if we consolidate terms, we get

$$= \frac{(4)(2)}{(2^{3/2})(2)^{1/2}} \times \frac{\pi}{\pi^{3/2}} \times \frac{m^{3/2}}{(m)(m)^{1/2}} \times \frac{E^{1/2}}{(E)^{1/2}} \times \frac{1}{(kT)^{3/2}} \times \exp\left(-\frac{E}{kT}\right)dE$$

$$= 2\sqrt{E/\pi}\left(\frac{1}{kT}\right)^{3/2}\exp\left(-E/kT\right)dE$$

and since we began by saying we wanted $f(v)\,dv = f_E(E)\,dE$, we have

$$f_E(E) = \frac{f(v)\,dv}{dE} = 2\sqrt{E/\pi}\left(\frac{1}{kT}\right)^{3/2}\exp\left(-E/kT\right) = f_\varepsilon$$

b) $\bar{E} = \langle E f_E \rangle = 2\sqrt{1/\pi}\left(\frac{1}{kT}\right)^{3/2}\underbrace{\int_0^\infty E^{3/2}\exp\left(-E/kT\right)dE}_{I}$

I is a gamma function integral, $\int_0^\infty x^{v-1}e^{-\alpha x}\,dx = \Gamma(v)/\alpha^v$

where in this case, $v = 5/2$ and $\alpha = \frac{1}{kT}$, so

$I = \Gamma(5/2)/(1/kT)^{5/2}$, then since $\Gamma\left(k+\frac{3}{2}\right) = \frac{(2k+1)!!}{2^{k+1}}\sqrt{\pi}$

$\int_0^\infty x^{n/2}e^{-\alpha x} = \frac{n!!\sqrt{\pi}}{2^{(n+1)/2}\alpha^{(n+2)/2}}$, where $n!! = n(n-2)(n-4)\ldots(1)$

(for an odd integer n)

$$I = \frac{3!!\sqrt{\pi}}{2^2\alpha^{5/2}} = \frac{3\sqrt{\pi}}{4(1/kT)^{5/2}}$$

$$\langle Ef_E \rangle = 2\sqrt{1/\pi}\left(\frac{1}{kT}\right)^{3/2}\frac{3\sqrt{\pi}}{4(1/kT)^{5/2}} = \frac{3}{2}kT = \bar{E} = \langle Ef_E \rangle$$

which we can plug back into our original expression.

$\bar{E}_m$ occurs for the value of E such that $df_E(E)/dE = 0$ for a given temperature T.

$$\frac{df_E}{dE} = \frac{d}{dE}\left(2\sqrt{E/\pi}\,(1/KT)^{3/2}\exp(-E/KT)\right)$$

$$= 2(1/KT)^{3/2}\sqrt{1/\pi}\,\frac{d}{dE}\left(\sqrt{E}\exp(-E/KT)\right)$$

$$= 2(1/KT)^{3/2}\sqrt{1/\pi}\left(\tfrac{1}{2}(E)^{-1/2}\exp(-E/KT) + \sqrt{E}\left(-\tfrac{1}{KT}\right)\exp(-E/KT)\right)$$

$$= 2(1/KT)^{3/2}\sqrt{1/\pi}\exp(-E/KT)\left(\tfrac{1}{2}E^{-1/2} - \sqrt{E}/KT\right)$$

Since we want $\frac{df_E}{dE} = 0$, this occurs only when $\frac{1}{2}E^{-1/2} - E^{1/2}/KT = 0$

so, we multiply through by $E^{1/2}$ to get $\frac{1}{2} - E/KT = 0 \Rightarrow E = KT/2$

which means $E_m = KT/2$

5. $f(v) = \left(\frac{\beta}{\pi}\right)^{3/2} e^{-\beta v^2}\left[1 + A v_x v_y + B v_z\left(v^2 - \frac{5}{2\beta}\right)\right]$

$m\langle \vec{v} f\rangle = m\left(\langle v_x f\rangle \hat{x} + \langle v_y f\rangle \hat{y} + \langle v_z f\rangle \hat{z}\right)$ take each component and solve independently.

$$\langle v_x f\rangle = \left(\frac{\beta}{\pi}\right)^{3/2}\iiint d\vec{v}\; e^{-\beta v^2}\left[v_x + A v_x^2 v_y + B v_x v_z\left(v^2 - \frac{5}{2\beta}\right)\right]$$
$$= \left(\frac{\beta}{\pi}\right)^{3/2}\left[\iiint_{\infty}^{\infty} v_x e^{-\beta v^2} d\vec{v} + \iiint_{\infty}^{\infty} A v_x^2 v_y e^{-\beta v^2} d\vec{v} + \iiint_{\infty}^{\infty} B v_x v_z\left(v^2 - \frac{5}{2\beta}\right) e^{-\beta v^2} d\vec{v}\right]$$

$$\iiint_{\infty} v_x e^{-\beta v^2} d\vec{v} = \int_{-\infty}^{\infty} e^{-\beta v_z^2} dv_z \int_{-\infty}^{\infty} e^{-\beta v_y^2} dv_y \int_{-\infty}^{\infty} v_x e^{-\beta v_x^2} dv_x$$
$$= 0 \quad (\text{Since } v_x e^{-\beta v_x^2} \text{ is an odd function})$$

$$\iiint_{\infty} A v_x^2 v_y e^{-\beta v^2} d\vec{v} = A\int_{-\infty}^{\infty} e^{-\beta v_z^2} dv_z \int_{-\infty}^{\infty} v_y e^{-\beta v_y^2} dv_y \int_{-\infty}^{\infty} v_x^2 e^{-\beta v_x^2} dv_x$$
$$= 0 \quad (\text{Since } v_y e^{-\beta v_y^2} \text{ is an odd function})$$

$$\iiint_{\infty} B v_x v_z\left(v_x^2 + v_y^2 + v_z^2 - \frac{5}{2\beta}\right) e^{-\beta v^2} d\vec{v}$$
$$= B\Bigg[\underbrace{\int_{-\infty}^{\infty} v_z e^{-\beta v_z^2} dv_z \int_{-\infty}^{\infty} e^{-\beta v_y^2} dv_y \int_{-\infty}^{\infty} v_x^3 e^{-\beta v_x^2} dv_x}_{=0 \text{ because of the two odd terms.}} +$$
$$\underbrace{\int_{-\infty}^{\infty} v_z e^{-\beta v_z^2} dv_z \int_{-\infty}^{\infty} v_y^2 e^{-\beta v_y^2} dv_y \int_{-\infty}^{\infty} v_x e^{-\beta v_x^2} dv_x}_{=0 \text{ because of the two odd terms.}} +$$
$$\underbrace{\int_{-\infty}^{\infty} v_z^3 e^{-\beta v_z^2} dv_z \int_{-\infty}^{\infty} e^{-\beta v_y^2} dv_y \int_{-\infty}^{\infty} v_x e^{-\beta v_x^2} dv_x}_{=0 \text{ because of the two odd terms.}} +$$
$$\underbrace{\left(-\frac{5}{2}\beta\right)\int_{-\infty}^{\infty} v_z e^{-\beta v_z^2} dv_z \int_{-\infty}^{\infty} e^{-\beta v_y^2} dv_y \int_{-\infty}^{\infty} v_x e^{-\beta v_x^2} dv_x}_{=0 \text{ because of the two odd terms.}}\Bigg] = 0$$

$$\langle v_x f\rangle = \left(\frac{\beta}{\pi}\right)^{3/2}[0 + 0 + 0] = 0$$

the whole process is explicitly shown for the x-component of velocity, but for the other components, the work of specifically expanding to throw out terms will be skipped, and the expression will immediately be written in terms of non-odd terms.

$$\langle v_y f \rangle = \left(\frac{\beta}{\pi}\right)^{3/2} \iiint_{\infty} d\vec{v}\; v_y\, e^{-\beta v^2}\left[1 + A v_x v_y + B v_z\left(v^2 - \frac{5}{2\beta}\right)\right]$$

$$= \left(\frac{\beta}{\pi}\right)^{3/2}\left[0 + 0 + 0\right] = 0$$

odd in y · odd in x · odd in y,z

$$\langle v_z f \rangle = \left(\frac{\beta}{\pi}\right)^{3/2} \iiint_{\infty} d\vec{v}\; v_z\, e^{-\beta v^2}\left[1 + A v_x v_y + B v_z\left(v^2 - \frac{5}{2\beta}\right)\right]$$

$$= \left(\frac{\beta}{\pi}\right)^{3/2}\left[0 + 0 + B \iiint_{\infty} d\vec{v}\; v_z^2\, e^{-\beta v^2}\left(v^2 - \frac{5}{2\beta}\right)\right]$$

odd in z · odd in x,y

$$= \left(\frac{\beta}{\pi}\right)^{3/2} B \underbrace{\iiint_{\infty} d\vec{v}\; v_z^2\, e^{-\beta v^2}\left(v^2 - \frac{5}{2\beta}\right)}_{I_{\langle v_z f\rangle}}$$

$$I_{\langle v_z f\rangle} = \int_{-\infty}^{\infty} v_z^2 e^{-\beta v_z^2} dv_z \int_{-\infty}^{\infty} e^{-\beta v_y^2} dv_y \int_{-\infty}^{\infty} v_x^2 e^{-\beta v_x^2} dv_x +$$
$$\int_{-\infty}^{\infty} v_z^2 e^{-\beta v_z^2} dv_z \int_{-\infty}^{\infty} v_y^2 e^{-\beta v_y^2} dv_y \int_{-\infty}^{\infty} e^{-\beta v_x^2} dv_x +$$
$$\int_{-\infty}^{\infty} v_z^4 e^{-\beta v_z^2} dv_z \int_{-\infty}^{\infty} e^{-\beta v_y^2} dv_y \int_{-\infty}^{\infty} e^{-\beta v_x^2} dv_x -$$
$$\frac{5}{2\beta} \int_{-\infty}^{\infty} v_z^2 e^{-\beta v_z^2} dv_z \int_{-\infty}^{\infty} e^{-\beta v_y^2} dv_y \int_{-\infty}^{\infty} e^{-\beta v_x^2} dv_x$$

there are recurring results worth defining for use throughout this question...

for $I_n(\beta) = \int_{-\infty}^{\infty} t^n e^{-t^2}$ we have the following results

1. $I_0(\beta) = \sqrt{\frac{\pi}{\beta}} = I_0$
2. $I_2(\beta) = \frac{\sqrt{\pi}}{2\beta^{3/2}} = I_2$
3. $I_4(\beta) = \frac{3\sqrt{\pi}}{4\beta^{5/2}} = I_4$

$$I_0 \times I_2^2 = \frac{\sqrt{\pi}}{2\beta^{3/2}} \frac{\sqrt{\pi}}{2\beta^{3/2}} \frac{\sqrt{\pi}}{\sqrt{\beta}} = \frac{\pi^{3/2}}{4\beta^{7/2}}$$

$$I_0^2 \times I_4 = \frac{\sqrt{\pi}}{\sqrt{\beta}} \frac{\sqrt{\pi}}{\sqrt{\beta}} \frac{3\sqrt{\pi}}{4\beta^{5/2}} = \frac{3}{4}\frac{\pi^{3/2}}{\beta^{7/2}}$$

$$I_0^2 \times I_2 = \frac{\sqrt{\pi}}{\sqrt{\beta}} \frac{\sqrt{\pi}}{\sqrt{\beta}} \frac{\sqrt{\pi}}{2\beta^{3/2}} = \frac{\pi^{3/2}}{2\beta^{5/2}}$$

utilizing these results gives...

$$I_{\langle v_z f\rangle} = I_0 \times I_2^2 + I_0 \times I_2^2 + I_0^2 \times I_4 - \frac{5}{2\beta} I_0^2 \times I_2$$

$$= \frac{(\pi)^{3/2}}{4\beta^{7/2}} + \frac{(\pi)^{3/2}}{4\beta^{7/2}} + \frac{3}{4}\frac{\pi^{3/2}}{\beta^{7/2}} - \frac{5}{2\beta}\cdot\frac{\pi^{3/2}}{2\beta^{5/2}}$$

$$= \frac{5}{4}\frac{\pi^{3/2}}{\beta^{7/2}} - \frac{5}{4}\frac{\pi^{3/2}}{\beta^{7/2}} = 0$$

So we have... $\langle v_x f\rangle = \langle v_y f\rangle = \langle v_z f\rangle = 0$

this means that $m\langle \vec{v} f \rangle = m\left(0\hat{x} + 0\hat{y} + 0\hat{z}\right) = 0$

Next: $m\langle v_x^2/2\, f\rangle$ for ...

$$f(v) = \left(\frac{\beta}{\pi}\right)^{3/2} e^{-\beta v^2}\left[1 + A v_x v_y + B v_z\left(v^2 - \frac{5}{2\beta}\right)\right]$$

$$m\left\langle \tfrac{1}{2} v_x^2 f\right\rangle = \tfrac{1}{2} m\left\langle v_x^2 f\right\rangle = \tfrac{1}{2} m\left(\int_{-\infty}^{\infty}\int\int d\vec{v}\; v_x^2\left(\frac{\beta}{\pi}\right)^{3/2} e^{-\beta v^2}\left[1 + A v_x v_y + B v_z\left(v^2 - \frac{5}{2\beta}\right)\right]\right)$$

which we can further simplify to...

$$\langle v_x^2 f\rangle = \left(\frac{\beta}{\pi}\right)^{3/2} \int_{-\infty}^{\infty}\int\int v_x^2 e^{-\beta v^2}\left[1 + \underbrace{A v_x v_y}_{\text{(odd terms)}} + \underbrace{B v_z\left(v^2 - 5/2\beta\right)}_{\text{(odd terms)}}\right] d\vec{v}$$

$$= \left(\frac{\beta}{\pi}\right)^{3/2} \int_{-\infty}^{\infty}\int\int v_x^2 e^{-\beta v^2} d\vec{v}$$

$$= \left(\frac{\beta}{\pi}\right)^{3/2} \int_{-\infty}^{\infty} e^{-\beta v_z^2} dv_z \int_{-\infty}^{\infty} e^{-\beta v_y^2} dv_y \int_{-\infty}^{\infty} v_x^2 e^{-\beta v_x^2} dv_x$$

$$= \left(\frac{\beta}{\pi}\right)^{3/2} I_0 \times I_0 \times I_2 = \frac{1}{2\beta} \quad \text{(using results from above)}$$

$$m\left\langle v_x^2/2\, f\right\rangle = \tfrac{1}{2} m\left\langle v_x^2 f\right\rangle = \tfrac{1}{2} m \frac{1}{2\beta} = \frac{m}{4\beta} = m\left\langle \frac{v_x^2}{2} f\right\rangle$$

Next: $m\langle v_y^2/2\, f\rangle$ for...

$$f(v) = \left(\frac{\beta}{\pi}\right)^{3/2} e^{-\beta v^2}\left[1 + A v_x v_y + B v_z\left(v^2 - \frac{5}{2\beta}\right)\right]$$

$$m\left\langle \tfrac{1}{2} v_y^2 f\right\rangle = \tfrac{1}{2} m\left\langle v_y^2 f\right\rangle = \tfrac{1}{2} m\left(\int_{-\infty}^{\infty}\int\int d\vec{v}\; v_y^2\left(\frac{\beta}{\pi}\right)^{3/2} e^{-\beta v^2}\left[1 + A v_x v_y + B v_z\left(v^2 - \frac{5}{2\beta}\right)\right]\right)$$

which we can further simplify to...

$$\langle v_x^2 f\rangle = \left(\frac{\beta}{\pi}\right)^{3/2} \int_{-\infty}^{\infty}\int\int v_y^2 e^{-\beta v^2}\left[1 + \underbrace{A v_x v_y}_{\text{(odd terms)}} + \underbrace{B v_z\left(v^2 - 5/2\beta\right)}_{\text{(odd terms)}}\right] d\vec{v}$$

$$= \left(\frac{\beta}{\pi}\right)^{3/2} \int_{-\infty}^{\infty}\int\int v_y^2 e^{-\beta v^2} d\vec{v}$$

$$= \left(\frac{\beta}{\pi}\right)^{3/2} \int_{-\infty}^{\infty} e^{-\beta v_z^2} dv_z \int_{-\infty}^{\infty} v_y^2 e^{-\beta v_y^2} dv_y \int_{-\infty}^{\infty} e^{-\beta v_z^2} dv_z$$

$$= \left(\frac{\beta}{\pi}\right)^{3/2} I_0 \times I_2 \times I_0 = \frac{1}{2\beta} \quad \text{(using results from above)}$$

$$m\langle v_y^2/2\, f\rangle = \tfrac{1}{2} m \langle v_y^2 f\rangle = \tfrac{1}{2} m \tfrac{1}{2\beta} = \frac{m}{4\beta} = m\left\langle \frac{v_y^2}{2} f\right\rangle$$

Next: $m\langle v_z^2/2\, f\rangle$ for...

$$f(v) = \left(\frac{\beta}{\pi}\right)^{3/2} e^{-\beta v^2}\left[1 + A v_x v_y + B v_z\left(v^2 - \frac{5}{2\beta}\right)\right]$$

$$m\left\langle \tfrac{1}{2} v_z^2 f\right\rangle = \tfrac{1}{2} m \langle v_z^2 f\rangle = \tfrac{1}{2} m \left(\int_\infty\int_\infty\int_\infty d\vec{v}\; v_z^2 \left(\frac{\beta}{\pi}\right)^{3/2} e^{-\beta v^2}\left[1 + A v_x v_y + B v_z\left(v^2 - \frac{5}{2\beta}\right)\right]\right)$$

which we can further simplify to...

$$\langle v_x^2 f\rangle = \left(\frac{\beta}{\pi}\right)^{3/2} \int_\infty\int_\infty\int_\infty v_z^2 e^{-\beta v^2}\left[1 + \cancel{A v_x v_y}^{\text{(odd terms)}} + \cancel{B v_z\left(v^2 - 5/2\beta\right)}^{\text{(odd terms)}}\right] d\vec{v}$$

$$= \left(\frac{\beta}{\pi}\right)^{3/2} \int_\infty\int_\infty\int_\infty v_z^2 e^{-\beta v^2} d\vec{v}$$

$$= \left(\frac{\beta}{\pi}\right)^{3/2} \int_{-\infty}^{\infty} v_z^2\, e^{-\beta v_z^2} dv_z \int_{-\infty}^{\infty} e^{-\beta v_y^2} dv_y \int_{-\infty}^{\infty} e^{-\beta v_z^2} dv_z$$

$$= \left(\frac{\beta}{\pi}\right)^{3/2} I_2 \times I_0 \times I_0 = \frac{1}{2\beta} \quad \text{(using results from above)}$$

$$m\langle v_z^2/2\, f\rangle = \tfrac{1}{2} m \langle v_z^2 f\rangle = \tfrac{1}{2} m \tfrac{1}{2\beta} = \frac{m}{4\beta} = m\left\langle \frac{v_z^2}{2} f\right\rangle$$

Next: $m\langle \vec{v}\,(v^2/2) f\rangle = \tfrac{1}{2} m \langle \vec{v}\,(v^2) f\rangle$ for...

$$f(v) = \left(\frac{\beta}{\pi}\right)^{3/2} e^{-\beta v^2}\left[1 + A v_x v_y + B v_z\left(v^2 - \frac{5}{2\beta}\right)\right]$$

$$\langle \vec{v}\, v^2 f\rangle = \langle v_x v^2 f\rangle \hat{x} + \langle v_y v^2 f\rangle \hat{y} + \langle v_z v^2 f\rangle \hat{z}$$

so solve one at a time...

$$\langle v_x v^2 f\rangle = \langle v_x^3 f\rangle + \langle v_x v_y^2 f\rangle + \langle v_x v_z^2 f\rangle$$

(when we multiply through by $v_x^3 e^{-\beta v^2}$)

$$\langle v_x^3 f\rangle = \left(\frac{\beta}{\pi}\right)^{3/2} \int_\infty\int_\infty\int_\infty v_x^3 e^{-\beta v^2}\left[1 + \cancel{A v_x v_y}^{\text{odd terms } (v_z)} + \cancel{B v_z\left(v^2 - \frac{5}{2\beta}\right)}^{\text{odd terms } (v_y)}\right] d\vec{v}$$

$$= 0$$

$$\langle V_x V_y^2 f\rangle = \left(\frac{\beta}{\pi}\right)^{\frac{3}{2}} \iiint_{\infty} V_x V_y^2 e^{-\beta v^2}\left[1 + A v_x v_y + B v_z\left(v^2 - \frac{5}{2\beta}\right)\right] d\vec{v}$$

$= 0$

odd $v_x e^{-\beta v_x^2}$ · odd $v_y^3 e^{-\beta v_y^2}$ · odd $v_x e^{-\beta v_x^2}$ and the v^2 after will just make it v_x^3 or leave it as v_x for each term.

$$\langle V_x V_z^2 f\rangle = \left(\frac{\beta}{\pi}\right)^{\frac{3}{2}} \iiint_{\infty} V_x V_z^2 e^{-\beta v^2}\left[1 + A v_x v_y + B v_z\left(v^2 - \frac{5}{2\beta}\right)\right] d\vec{v}$$

$= 0$

odd $v_x e^{-\beta v_x^2}$ · odd $v_y e^{-\beta v_y^2}$ · odd v_x, v_z

So... $\langle V_x V^2 f\rangle = 0 + 0 + 0 = 0$

Next, solve...

$$\langle V_y V^2 f\rangle = \langle V_y V_x^2 f\rangle + \langle V_y^3 f\rangle + \langle V_y V_z^2 f\rangle$$

$$\langle V_y V_x^2 f\rangle = \left(\frac{\beta}{\pi}\right)^{\frac{3}{2}} \iiint_{\infty} V_y V_x^2 e^{-\beta v^2}\left[1 + A v_x v_y + B v_z\left(v^2 - \frac{5}{2\beta}\right)\right] d\vec{v}$$

$= 0$

odd $v_y e^{-\beta v_y^2}$ · odd $v_x^3 e^{-\beta v_x^2}$ · odd v_y, v_z

$$\langle V_y^3 f\rangle = \left(\frac{\beta}{\pi}\right)^{\frac{3}{2}} \iiint_{\infty} V_y^3 e^{-\beta v^2}\left[1 + A v_x v_y + B v_z\left(v^2 - \frac{5}{2\beta}\right)\right] d\vec{v}$$

$= 0$

odd $v_y^3 e^{-\beta v_y^2}$ · odd $v_x e^{-\beta v_x^2}$ · odd v_y, v_z

$$\langle V_y V_z^2 f\rangle = \left(\frac{\beta}{\pi}\right)^{\frac{3}{2}} \iiint_{\infty} V_y V_z^2 e^{-\beta v^2}\left[1 + A v_x v_y + B v_z\left(v^2 - \frac{5}{2\beta}\right)\right] d\vec{v}$$

$= 0$

odd $v_y e^{-\beta v_y^2}$ · odd $v_x e^{-\beta v_x^2}$ · odd v_y, v_z

$\langle V_y V^2 f\rangle = 0 + 0 + 0$

Next solve...

$$\langle V_z V^2 f\rangle = \langle V_z V_x^2 f\rangle + \langle V_z V_y^2 f\rangle + \langle V_z^3 f\rangle$$

$$\langle V_z V_x^2 f\rangle = \left(\frac{\beta}{\pi}\right)^{3/2}\iiint_{\infty} V_z V_x^2 e^{-\beta V^2}\left[\cancel{1}^{\,0} + \cancel{A V_x V_y}^{\,0} + BV_z\left(V^2 - \frac{5}{2\beta}\right)\right] d\vec{V}$$

odd $V_z e^{-\beta V_z^2}$; odd $V_y e^{-\beta V_y^2}$

$$= \left(\frac{\beta}{\pi}\right)^{3/2}\iiint_{\infty} B\, V_z^2 V_x^2 e^{-\beta V^2}\left(V^2 - \frac{5}{2\beta}\right) d\vec{V}$$

$$= \left(\frac{\beta}{\pi}\right)^{3/2}\left[\iiint_{\infty} B V_z^2 V_x^4 e^{-\beta V^2} d\vec{V} + \iiint_{\infty} B V_z^2 V_x^2 V_y^2 e^{-\beta V^2} d\vec{V}\right.$$

$$\left. + \iiint_{\infty} B V_z^4 V_x^2 e^{-\beta V^2} d\vec{V} - \frac{5}{2\beta}\iiint_{\infty} B V_z^2 V_x^2 e^{-\beta V^2} dV\right]$$

$$= \left(\frac{\beta}{\pi}\right)^{3/2} B\left(I_0 I_2 I_4 + I_2^3 + I_0 I_2 I_4 - I_2 I_0 I_2 \frac{5}{2\beta}\right)$$

$$= \frac{B}{4\beta^3}$$

$$\langle V_z V_y^2 f\rangle = \left(\frac{\beta}{\pi}\right)^{3/2}\iiint_{\infty} V_z V_y^2 e^{-\beta V^2}\left[\cancel{1}^{\,0} + \cancel{A V_x V_y}^{\,0} + BV_z\left(V^2 - \frac{5}{2\beta}\right)\right] d\vec{V}$$

odd $V_z e^{-\beta V_z^2}$; odd $V_y^3 e^{-\beta V_y^2}$ and $V_x e^{-\beta V_x^2}$

we can immediately see that $\langle V_z V_y^2 f\rangle = \langle V_z V_x^2 f\rangle = \frac{B}{4\beta^3}$

$$\langle V_z^3 f\rangle = \left(\frac{\beta}{\pi}\right)^{3/2}\iiint_{\infty} V_z^3 e^{-\beta V^2}\left[\cancel{1}^{\,0} + \cancel{A V_x V_y}^{\,0} + BV_z\left(V^2 - \frac{5}{2\beta}\right)\right] d\vec{V}$$

odd $V_z^3 e^{-\beta V_z^2}$; odd V_x, V_y, V_z

$$= \left(\frac{\beta}{\pi}\right)^{3/2}\iiint_{\infty} V_z^4 e^{-\beta V^2} B\left(V^2 - \frac{5}{2\beta}\right)\Big] d\vec{V}$$

$$= \left(\frac{\beta}{\pi}\right)^{3/2} B\left[\iiint_{\infty} V_z^4 V_x^2 e^{-\beta V^2} d\vec{V} + \iiint_{\infty} V_z^4 V_y^2 e^{-\beta V^2} d\vec{V}\right.$$

$$\left. + \iiint_{\infty} V_z^6 e^{-\beta V^2} d\vec{V} - \frac{5}{2\beta}\iiint_{\infty} V_z^4 e^{-\beta V^2} d\vec{V}\right]$$

So we need to define a new quantity

$$I_6 = \int_{-\infty}^{\infty} t^6 e^{-\beta t^2} dt = \frac{15\sqrt{\pi}}{8\beta^{7/2}}$$

$$\langle V_z^3 f\rangle = \left(\frac{\beta}{\pi}\right)^{3/2} B\left[I_4 I_2 I_0 + I_4 I_2 I_0 + I_6 I_0 I_0 - \frac{5}{2\beta} I_4 I_0 I_0\right]$$

$$= \frac{3B}{4\beta^3}$$

$$\langle V_z V^2 f\rangle = \frac{5B}{4\beta^3}$$

$$\langle \vec{v}\, V^2 f\rangle = \langle V_x V^2 f\rangle \hat{x} + \langle V_y V^2 f\rangle \hat{y} + \langle V_z V^2 f\rangle \hat{z} = 0\hat{x} + 0\hat{y} + \frac{5B}{4\beta^3}\hat{z}$$

$$= \frac{5B}{4\beta^3}\hat{z}$$

$$m\langle \vec{v}\,(v^2/2)f\rangle = \frac{1}{2}m\langle \vec{v}\,(v^2)f\rangle = \frac{1}{2}m\frac{5B}{4\beta^3}\hat{z} = \frac{5Bm}{8\beta^3}\hat{z}$$

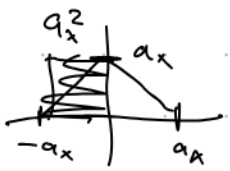
$$\boxed{m\langle \vec{v}\,(v^2/2)f\rangle = \frac{5Bm}{8\beta^3}\hat{z}}$$

6. non-equilibrium, spatially uniform, monoatomic

$$F(\vec{v}) = C\, g_x(v_x)\, g_y(v_y)\, g_z(v_z)$$

a) given number density n, determine the normalization factor C

$$f(\vec{v}) = F(\vec{v})/n = \frac{C}{n}\, g_x(v_x)\, g_y(v_y)\, g_z(v_z)$$



$$\iiint_\infty f(\vec{v})\,d\vec{V} = \iiint_\infty \frac{C}{n}\, g_x(v_x)\, g_y(v_y)\, g_z(v_z)\,d\vec{V} = 1$$

$$= \frac{C}{n}\left[\underbrace{\int_{-a_x}^{a_x} g_x(v_x)\,dv_x}_{a_x^2}\ \underbrace{\int_{-a_y}^{a_y} g_y(v_y)\,dv_y}_{a_y^2}\ \underbrace{\int_{-a_z}^{a_z} g_z(v_z)\,dv_z}_{a_z^2}\right]$$

$$1 = \frac{C}{n}\, a_x^2 a_y^2 a_z^2 \Rightarrow \boxed{C = \frac{n}{a_x^2 a_y^2 a_z^2}}$$

$$\text{So } f(\vec{v}) = \frac{1}{a_x^2 a_y^2 a_z^2} \iiint_\infty g_x(v_x)\, g_y(v_y)\, g_z(v_z)\,d\vec{V}$$

b) $$\vec{\bar{v}} = \langle \vec{v} f(\vec{v}) \rangle = \langle v_x f(\vec{v}) \rangle \hat{x} + \langle v_y f(\vec{v}) \rangle \hat{y} + \langle v_z f(\vec{v}) \rangle \hat{z}$$

$$\langle v_x f(\vec{v}) \rangle = \frac{1}{a_x^2 a_y^2 a_z^2} \iiint_\infty v_x\, g_x(v_x)\, g_y(v_y)\, g_z(v_z)\,d\vec{V}$$

$$= \frac{1}{a_x^2 a_y^2 a_z^2}\left(\int_{-a_x}^{0} v_x(a_x+v_x)\,dv_x + \int_0^{a_x} v_x(a_x - v_x)\,dv_x\right) \underbrace{\int_{-a_y}^{a_y} (a_y - v_y)\,dv_y}_{a_y^2} \underbrace{\int_{-a_z}^{a_z} (a_z - v_z)\,dv_z}_{a_z^2}$$

$$= \frac{1}{a_x^2 a_y^2 a_z^2}\left(-\int_0^{-a_x} (v_x a_x + v_x^2)\,dv_x + \int_0^{a_x} (v_x a_x - v_x^2)\,dv_x\right) a_y^2 a_z^2$$

$$\int_a^b (v_x a_x + v_x^2)\,dv_x = \left[a_x \frac{v_x^2}{2} + \frac{v_x^3}{3}\right]_a^b, \qquad \int_a^b (v_x a_x - v_x^2)\,dv_x = \left[a_x \frac{v_x^2}{2} - \frac{v_x^3}{3}\right]_a^b$$

$$\langle v_x f(\vec{v}) \rangle = \frac{\cancel{a_y^2}\,\cancel{a_z^2}}{a_x^2 \cancel{a_y^2}\,\cancel{a_z^2}}\left(-\int_0^{-a_x} (v_x a_x + v_x^2)\,dv_x + \int_0^{a_x} (v_x a_x - v_x^2)\,dv_x\right)$$

$$= \frac{1}{a_x^2}\left(-\left[a_x \frac{v_x^2}{2} + \frac{v_x^3}{3}\right]_0^{-a_x} + \left[a_x \frac{v_x^2}{2} - a_x \frac{v_x^2}{2}\right]_0^{a_x}\right)$$

$$= \frac{1}{a_x^2}\left(-\cancel{\frac{a_x^3}{2}} + \cancel{\frac{a_x^3}{3}} + \cancel{\frac{a_x^3}{2}} - \cancel{\frac{a_x^3}{3}}\right) = 0$$

$\langle V_x f(\vec{v}) \rangle = 0$

then for the other velocity directions, just change indices

$\langle V_z f(\vec{v}) \rangle = 0$ $\quad$ $\langle V_z f(\vec{v}) \rangle = 0$

$$\vec{\bar{v}} = \langle \vec{v} f(\vec{v}) \rangle = \langle V_x f(\vec{v}) \rangle \hat{x} + \langle V_y f(\vec{v}) \rangle \hat{y} + \langle V_z f(\vec{v}) \rangle \hat{z}$$
$$\vec{\bar{v}} = 0\hat{x} + 0\hat{y} + 0\hat{z} = \vec{0}$$

c) the pressure tensor P_{ij} can be obtained by the moment $\langle m v_i v_j F \rangle = \rho u_i u_j + P_{ij}$, but from b), we know $u_i = u_j = 0 \; \forall \; i,j \in \{1,2,3\}$

so for this specific NDF F_1, $\langle m v_i v_j F \rangle = P_{ij}$

$$m \langle v_i v_j F \rangle = \frac{mn}{a_x^2 a_y^2 a_z^2} \iiint_{\infty} v_i v_j \, g_x(v_x) g_y(v_y) \, g_z(v_z) \, dv_k$$

$$= \frac{\rho}{a_x^2 a_y^2 a_z^2} \int_{-a_x}^{a_x} \int_{-a_y}^{a_y} \int_{-a_z}^{a_z} V_i V_j \, g_x(v_x) g_y(v_y) \, g_z(v_z) \, dv_z \, dv_y \, dv_x$$

split into 2 cases: ① $\rightarrow i = j$ ② $\rightarrow i \neq j$ $\quad (i,j \in \{\overset{x}{1}, \overset{y}{2}, \overset{z}{3}\})$

for case ①, assume $i = j = 1$, so we are solving ...

$$m \langle V_x V_x F \rangle = \frac{\rho}{a_x^2 a_y^2 a_z^2} \int_{-a_x}^{a_x} V_x^2 \, g_x(v_x) \, dv_x \underbrace{\int_{-a_y}^{a_y} g_y(v_y) dv_y}_{a_y^2} \underbrace{\int_{-a_z}^{a_z} g_z(v_z) dv_z}_{a_z^2}$$

$$= \frac{\rho \cancel{a_y^2} \cancel{a_z^2}}{a_x^2 \cancel{a_y^2} \cancel{a_z^2}} \left[\int_0^{a_x} V_x^2 (a_x - V_x) \, dV_x - \int_0^{-a_x} V_x^2 (a_x + V_x) dV_x \right]$$

$$= \rho / a_x^2 \left[\int_0^{a_x} (V_x^2 a_x - V_x^3) dV_x - \int_0^{-a_x} (V_x^2 a_x + V_x^3) dV_x \right]$$

$$= \rho/a_x^2 \left(\left[\frac{V_x^3}{3} a_x - \frac{V_x^4}{4} \right]_0^{a_x} - \left[\frac{V_x^3}{3} a_x + \frac{V_x^4}{4} \right]_0^{-a_x} \right)$$
$$= \rho/a_x^2 \left(\frac{a_x^4}{3} - \frac{a_x^4}{4} + \frac{a_x^4}{3} - \frac{a_x^4}{4} \right) = \frac{\rho}{a_x^2} \left(\frac{1}{6} a_x^4 \right) = \frac{\rho a_x^2}{6}$$

$m \langle V_x V_x F \rangle = \frac{\rho a_x^2}{6}$, so since in this problem we can freely rotate indices, we see that

$$m \langle V_i V_j F \rangle = \frac{\rho a_i^2}{6} \quad \text{for } i = j$$

for case (2), assume $i = 1$, $j = 2$, so we are solving

$$m \langle V_x V_y F \rangle = \frac{\rho}{a_x^2 a_y^2 a_z^2} \underbrace{\int_{-a}^{a} V_x g_x(V_x) dV_x}_{0} \underbrace{\int_{-a}^{a} V_y g_y(V_y) dV_y}_{0} \underbrace{\int_{-a}^{a} g_z(V_z) dV_z}_{a_z^2}$$

from the results shown in b), we see that $m \langle V_x V_y F \rangle = 0$, and again due to the form of F, we can switch indices of our velocities, and this applies for all off diagonal terms.

$$m \langle V_i V_j F \rangle = 0 \quad \text{for } i \neq j$$

so we can write the whole pressure tensor in the following compact form

$$P_{ij} = \delta_{ij} \frac{\rho a_i^2}{6}$$, which corresponds to pressure ...

$$p = \frac{1}{3} P_{ii} = \frac{\rho}{18}\left(a_x^2 + a_y^2 + a_z^2\right)$$

d) we recover $\vec{q}$ by taking the moment...

$$\frac{m}{2}\langle v_i v^2 F\rangle = \rho u_i\left(\frac{1}{2}u^2 + \frac{5}{2}\frac{p}{\rho}\right) - \tau_{ij}u_j + q_i$$ where we know the random velocity $u_i = 0 \;\forall\; i \in \{1,2,3\}$, so we can simplify this to...

$$\frac{m}{2}\langle v_i v^2 F\rangle = q_i$$, which is the bulk energy flux.

$$\frac{m}{2}\langle v_i v^2 F\rangle = \frac{m}{2}\iiint_{\infty}^{\infty} \frac{n}{a_x^2 a_y^2 a_z^2}\, v_i v^2\, g_x(v_x)\, g_y(v_y)\, g_z(v_z)\, dv_i$$

first, solve for...

$$\frac{m}{2}\langle v_x v^2 F = \frac{\rho}{2a_x^2 a_y^2 a_z^2}\iiint_{\infty}^{\infty} v_x v^2\, g_x(v_x) g_y(v_y)\, g_z(v_z)\, dv_x dv_y dv_z$$

(triangular fcn. is even, v_x^3 is odd, so this symmetric integral is zero!)

$$= \frac{\rho}{2a_x^2 a_y^2 a_z^2}\Big[\int_{-a_x}^{a_x} v_x^3 g_x(v_x)\, dv_x \int_{-a_y}^{a_y} g_y(v_y) dv_y \int_{-a_z}^{a_z} g_z(v_z)\, dv_z +$$
$$\int_{-a_x}^{a_x} v_x g_x(v_x) dv_x \int_{-a_y}^{a_y} v_y^2 g_y(v_y)\, dv_y \int_{-a_z}^{a_z} g_z(v_z) dv_z +$$
$$\int_{-a_x}^{a_x} v_x g_x(v_x) dv_x \int_{-a_y}^{a_y} g_y(v_y) dv_y \int_{-a_z}^{a_z} v_z^2 g_z(v_z) dv_z\Big]$$

$$= 0$$

Since we can swap indices due to the form of F, we see that

$$\frac{m}{2}\langle \vec{v} v^2 F\rangle = \vec{q} = 0\hat{x} + 0\hat{y} + 0\hat{z} = \vec{0}$$

7. Boltzmann Equation: $\frac{\partial F}{\partial t} + \vec{v}\cdot\vec{\nabla}_x F + \vec{a}\cdot\vec{\nabla}_v F = \frac{\delta F}{\delta t}$

mathematically, galilean invariance requires that $Q' = Q$ where Q' is in the transformed frame, and Q is in the initial frame.

given... $x_i = x_i' + u_{0,i}t$, $t = t'$ (time is unchanged for Galilean transform)
$x_i' = x_i - u_{0,i}t$

$$v_i = \frac{\partial x_i}{\partial t} = \frac{\partial x_i'}{\partial t} + u_{0,i} \qquad v_i = v_i' + u_{0,i}$$

$$a_i = \frac{\partial v_i}{\partial t} = \frac{\partial v_i'}{\partial t} + 0 = a_i' \qquad a_i = a_i'$$

$$F'(\vec{x}',\vec{v}',t) = F(\vec{x}' + \vec{u}_0 t, \vec{v}' + \vec{u}_0, t') \quad \text{or} \quad F(\vec{x},\vec{v},t) = F'(\vec{x} - \vec{u}_0 t, \vec{v} - \vec{u}_0, t)$$

we also require the derivative properties of Lagrangian transforming w.r.t. F, using multivariable chain rule. (assuming coordinate directions of $\vec{x}$ and $\vec{v}$ are independent)

$$\frac{\partial F}{\partial x_i} = \frac{\partial F'}{\partial x_i'}\frac{\partial x_i'}{\partial x_i}^{1} + \frac{\partial F'}{\partial v_j'}\frac{\partial v_j'}{\partial x_i}^{0} + \frac{\partial F'}{\partial t'}\frac{\partial t'}{\partial x_i}^{0} = \frac{\partial F'}{\partial x_i'}$$

so as an operator $\vec{\nabla}_x = \vec{\nabla}_{x'}$

$$\frac{\partial F}{\partial v_i} = \frac{\partial F'}{\partial x_j'}\frac{\partial x_j'}{\partial v_i}^{0} + \frac{\partial F'}{\partial v_i'}\frac{\partial v_i'}{\partial v_i}^{1} + \frac{\partial F'}{\partial t'}\frac{\partial t'}{\partial v_i}^{0}$$

so as an operator, $\vec{\nabla}_v = \vec{\nabla}_{v'}$

$$\frac{\partial F}{\partial t} = \frac{\partial F'}{\partial x_i'}\frac{\partial x_i'}{\partial t} + \frac{\partial F'}{\partial v_i'}\frac{\partial v_i'}{\partial t}^{0} + \frac{\partial F'}{\partial t'}\frac{\partial t'}{\partial t}^{1} = -\frac{\partial F'}{\partial x_i'}u_{0,i} + \frac{\partial F'}{\partial t'}$$

($\frac{\partial x_i'}{\partial t} = -u_{0,i}$)

so as an operator, $\frac{\partial}{\partial t} = \frac{\partial}{\partial t} - \vec{u}_0\cdot\vec{\nabla}_{x'}$

Now for the Boltzmann collision operator: (here, i denotes an inverse collision (scattering in))

$$\frac{\delta F}{\delta t} = \iiint_{\infty} d^3v_2 \int_0^{2\pi} d\epsilon \int_0^{\pi} d\chi \sin(\chi) S(g,\chi) g \left[F^i F_2^i - F F_2\right]$$

$d_3v_2 = d(\vec{v}_2' + \vec{u}_0)$, $d\epsilon = d\epsilon'$, $d\chi = d\chi'$, $\sin(\chi) = \sin(\chi')$

$g = g'$ $(\vec{v}_2 - \vec{v}_1 = \vec{v}_2' + \vec{u}_0 - \vec{v}_1' - \vec{u}_0 = \vec{v}_2' - \vec{v}_1')$, $S(g,\chi) = S'(g',\chi')$

and from above...

$F'(\vec{x}',\vec{v}',t) = F(\vec{x}' + \vec{u}_0 t, \vec{v}' + \vec{u}_0, t')$, so sub everything in...

$$\frac{\delta F}{\delta t} = \iiint_{\infty} d(\vec{v}' + \vec{u}_0) \int_0^{2\pi} d\epsilon' \int_0^{\pi} dx' \sin(x') S'(g', x') g' \times$$
$$[F^i(\vec{x}' + \vec{u}_0 t, \vec{v}' + \vec{u}_0, t') F_2^i(\vec{x}' + \vec{u}_0 t, \vec{v}' + \vec{u}_0, t') -$$
$$F(\vec{x}' + \vec{u}_0 t, \vec{v}' + \vec{u}_0, t') F_2(\vec{x}' + \vec{u}_0 t, \vec{v}' + \vec{u}_0, t')]$$

we see that for the specific values of F^i, F_2^i, F, F_2 given above, we can rewrite these as NDFs in the transformed space at $\vec{x}', \vec{v}', t$.

$$\frac{\delta F}{\delta t} = \iiint_{\infty} d(\vec{v}' + \vec{u}_0) \int_0^{2\pi} d\epsilon' \int_0^{\pi} dx' \sin(x') S'(g', x') g' [F'^i F_2'^i - F' F_2']$$

then for the triple integral, instead of integrating over $(\vec{v} + \vec{u}_0)$, we can simply shift the limits of integration by $\vec{u}_0$ (through u-substitution), so the integral is now...

$$\frac{\delta F}{\delta t} = \iiint_{\infty} d\vec{v}' \int_0^{2\pi} d\epsilon' \int_0^{\pi} dx' \sin(x') S'(g', x') g' [F'^i F_2'^i - F' F_2']$$

which is just $\frac{\delta F'}{\delta t'}$

$$\therefore \frac{\delta F}{\delta t} = \frac{\delta F'}{\delta t'}$$

Now we apply these operators and coordinate transformations to the boltzmann equation.

$$\frac{\partial F'}{\partial t'} - \vec{u}_0 \cdot \vec{\nabla}_{x'} F' + (\vec{v}' + \vec{u}_0) \cdot \vec{\nabla}_{x'} F' + \vec{a}' \cdot \vec{\nabla}_{v'} F' = \frac{\delta F'}{\delta t'}$$
$$\frac{\partial F'}{\partial t'} - \cancel{\vec{u}_0 \cdot \vec{\nabla}_{x'} F'} + \cancel{\vec{u}_0 \cdot \vec{\nabla}_{x'} F'} + \vec{v}' \cdot \vec{\nabla}_{x'} F + \vec{a}' \cdot \vec{\nabla}_{v'} F' = \frac{\delta F'}{\delta t'}$$
$$\frac{\partial F'}{\partial t'} + \vec{v}' \cdot \vec{\nabla}_{x'} F' + \vec{a}' \cdot \vec{\nabla}_{v'} F' = \frac{\delta F'}{\delta t'}$$

so the equation is invariant for all Galilean transformations.

8. a) show equivalence

$$\frac{\partial F}{\partial t} + \underbrace{\vec{\nabla} \cdot (F\vec{v})}_{①} + \underbrace{\vec{\nabla}_v \cdot (F\vec{a})}_{②} = \frac{\delta F}{\delta t}$$

expanding ① gives... $v_i \frac{\partial F}{\partial x_i} + F \frac{\partial v_i}{\partial x_i} = v_i \frac{\partial F}{\partial x_i}$ ← position and velocity are indep. in phase space

expanding ② gives... $a_i \frac{\partial F}{\partial v_i} + F \frac{\partial a_i}{\partial v_i} = a_i \frac{\partial F}{\partial v_i}$

So substituting and rewriting the equation gives...

$$\boxed{\frac{\partial F}{\partial t} + v_i \frac{\partial F}{\partial x_i} + a_i \frac{\partial F}{\partial v_i} = \frac{\delta F}{\delta t}}$$

b) $\frac{\partial F}{\partial t} + v_i \frac{\partial F}{\partial x_i} + a_i \frac{\partial F}{\partial v_i} = \frac{\delta F}{\delta t}$ multiply by mc_j, and integrate over velo. space.

$$\iiint_\infty d\vec{v}\, mc_j \left(\frac{\partial F}{\partial t} + v_i \frac{\partial F}{\partial x_i} + a_i \frac{\partial F}{\partial v_i} \right) = \iiint_\infty mc_j \frac{\delta F}{\delta t} d\vec{v}$$

$$m\left[\underbrace{\iiint_\infty c_j \frac{\partial F}{\partial t} d\vec{v}}_{①} + \underbrace{\iiint_\infty c_j v_i \frac{\partial F}{\partial x_i} d\vec{v}}_{②} + \underbrace{\iiint_\infty c_j a_i \frac{\partial F}{\partial v_i} d\vec{v}}_{③} \right] = \underbrace{m \iiint_\infty c_j \frac{\delta F}{\delta t} d\vec{v}}_{④}$$

start with ①:

$$\left\langle c_j \frac{\partial F}{\partial t} \right\rangle = \left\langle \frac{\partial}{\partial t}(c_j F) - F \frac{\partial c_j}{\partial t} \right\rangle = \frac{\partial}{\partial t} \langle c_j F \rangle - \left\langle F \frac{\partial v_j}{\partial t} \right\rangle + \left\langle F \frac{\partial u_j}{\partial t} \right\rangle$$

$$= \frac{\partial u_j}{\partial t} \langle F \rangle = \frac{\partial u_j}{\partial t} n$$

now for ②:

$$\left\langle c_j v_i \frac{\partial F}{\partial x_i} \right\rangle = \left\langle \frac{\partial}{\partial x_i}(c_j v_i F) \right\rangle - \left\langle \frac{\partial c_j}{\partial x_i} v_i F \right\rangle - \left\langle c_j \frac{\partial v_i}{\partial x_i} F \right\rangle$$

$$= \frac{\partial}{\partial x_i} \langle c_j v_i F \rangle - \left\langle \frac{\partial v_j}{\partial x_i} v_i F \right\rangle + \left\langle \frac{\partial u_j}{\partial x_i} v_i F \right\rangle$$

$$= \frac{\partial}{\partial x_i} \langle c_j v_i F \rangle + \frac{\partial u_j}{\partial x_i} \langle v_i F \rangle$$

$$= \frac{\partial}{\partial x_i} \langle V_j V_i F \rangle - \frac{\partial}{\partial x_i}\left(u_j \langle v_i F \rangle\right) + \frac{\partial u_j}{\partial x_i} \langle v_i F \rangle$$

$$= \frac{\partial}{\partial x_i}\langle v_j v_i F\rangle - \cancel{\frac{\partial u_j}{\partial x_i}\langle v_i F\rangle} - u_j \frac{\partial \langle v_i F\rangle}{\partial x_i} + \cancel{\frac{\partial u_j}{\partial x_i}\langle v_i F\rangle}$$

$$= \frac{\partial}{\partial x_i}\langle v_j v_i F\rangle - u_j \frac{\partial}{\partial x_i}(u_i n)$$

$$= \frac{\partial}{\partial x_i}\left(n u_j u_i + \frac{P_{ji}}{m}\right) - u_j \frac{\partial}{\partial x_i}(u_i n)$$

$$= \cancel{\frac{\partial n}{\partial x_i} u_j u_i} + \cancel{n \frac{\partial u_j}{\partial x_i} u_i} + n u_j \frac{\partial u_i}{\partial x_i} + \frac{1}{m}\frac{\partial P_{ji}}{\partial x_i} - \cancel{u_j \frac{\partial u_i}{\partial x_i} n} - \cancel{u_j u_i \frac{\partial n}{\partial x_i}}$$

$$= n u_j \frac{\partial u_i}{\partial x_i} + \frac{1}{m}\frac{\partial P_{ji}}{\partial x_i}$$

now for ③:

$$\langle C_j a_i \frac{\partial F}{\partial v_i}\rangle = \frac{\partial}{\partial v_i}\langle C_j a_i F\rangle - \langle \frac{\partial C_j}{\partial v_i} a_i F\rangle - \cancel{\langle C_j \frac{\partial a_j}{\partial v_i} F\rangle}^{\,0}$$

$$= \frac{\partial}{\partial v_i}\langle C_j a_i F\rangle - \langle \cancel{\frac{\partial v_j}{\partial v_i}} a_i F\rangle + \cancel{\langle \frac{\partial u_j}{\partial v_i} a_i F\rangle}^{\,0}$$

$$= \frac{\partial}{\partial v_i} a_i \langle C_j F\rangle - \delta_{ij} a_i \langle F\rangle$$

$$= \frac{\partial}{\partial v_i} a_i \langle v_j F\rangle - \frac{\partial}{\partial v_i} a_i u_j \langle F\rangle - \delta_{ij} a_i \langle F\rangle$$

$$= \cancel{\frac{\partial}{\partial v_i} a_i u_j n} - \cancel{\frac{\partial}{\partial v_i} a_i u_j n} - \delta_{ij} a_i n$$

$$= -\delta_{ij} a_i n = -a_i n$$

now for ④:

$$m\langle C_j \frac{\delta F}{\delta t}\rangle = \underbrace{m\cancel{\langle v_j \frac{\delta F}{\delta t}\rangle}^{\,0} - u_j m \cancel{\langle \frac{\delta F}{\delta t}\rangle}^{\,0}}_{\text{invariants of the Boltzmann collision integral}} = 0$$

if we then plug the results above back into ①-④, we get...

$$m\left[\frac{\partial u_j}{\partial t} n + n u_j \frac{\partial u_i}{\partial x_i} + \frac{1}{m}\frac{\partial P_{ji}}{\partial x_i} - a_i n\right] = 0$$

$$\rho \frac{\partial u_j}{\partial t} + \rho u_j \frac{\partial u_i}{\partial x_i} + \frac{\partial P_{ji}}{\partial x_i} - a_i \rho = 0$$

$$\rho \frac{\partial u_j}{\partial t} + \rho u_j \frac{\partial u_i}{\partial x_i} + \frac{\partial}{\partial x_i}(p\,\delta_{ij} - \tau_{ij}) = a_i \rho$$

or, in vector notation...

$$\rho \frac{\partial \vec{u}}{\partial t} + \nabla \cdot \left(\rho \vec{u}\vec{u} + p\,\overleftrightarrow{I} - \overleftrightarrow{\tau}\right) = \rho \vec{a}$$